

SLA PoD Corrective Maintenance Analysis (LATIS)

Objective

You are required to build an Alteryx workflow that analyzes SLA compliance for Corrective Maintenance tickets under the LATIS sub-project. This task assesses your ability to clean data, apply logic, calculate SLA KPIs, and structure output summaries.

Files Provided

- Full PCM History (Excel) – Includes all fault tickets.
- Excluded Work Orders (Excel) – Contains work orders that must be excluded.

Business Logic Overview

LATIS CM tickets are evaluated against two SLA thresholds:

1. Target SLA: Standard SLA commitment
2. Hurdle SLA: Relaxed backup SLA target

You will build logic to determine if each ticket:

- Exceeded or met Target SLA
- Exceeded or met Hurdle SLA
- Compute SLA achievement rates by severity level

Instructions (Step-by-Step)

1. Data Preparation

- Import both Excel files.
- Remove any work orders listed in the Excluded Work Orders file using a Join or Filter.

2. Filters to Apply

- Status does NOT contain “Escalated” or “Cancel”
- Sub Project equals LATIS
- TT Type is in (EM, Reference_EM, Telecom, Reference_Telecom)
- Title does NOT contain: HC, PT:, PM Error, Performance Issue, Visibility, Dust, Chassis, Health

3. Field Transformations

- Convert required fields to V_WString
- Convert Created At and Fault Recovery Time to DateTime
- Calculate Duration (seconds) = Fault Recovery Time - Created At

4. NSA Impact Logic

- Create Affected? field:
if Contains([Impact], "NSA") then "NSA" else "SA" endif
- Create/Update Fault Level:
if [Affected?] = "NSA" then "Minor" else [Fault Level] endif

SLA Threshold Logic

A. Target SLA Violation (Field: Target)

```
if [Fault Level]="Emergency" AND [Duration]>7200 THEN "NOK"  
elseif [Fault Level]="Critical" AND [Duration]>14400 THEN "NOK"  
elseif [Fault Level]="Major" AND [Duration]>28800 THEN "NOK"  
elseif [Fault Level]="Minor" AND [Duration]>43200 THEN "NOK"  
elseif [Resolved Owner]=""" THEN "NOK"  
elseif [Fault Recovery Time]="N/A" or [Fault Recovery Time]=""" or isnull([Fault Recovery Time]) THEN "NOK"  
else "" endif
```

B. Hurdle SLA Violation (Field: Hurdle)

```
if [Fault Level]="Emergency" AND [Duration]>14400 THEN "NOK"  
elseif [Fault Level]="Critical" AND [Duration]>21600 THEN "NOK"  
elseif [Fault Level]="Major" AND [Duration]>43200 THEN "NOK"  
elseif [Fault Level]="Minor" AND [Duration]>86400 THEN "NOK"  
elseif [Resolved Owner]=""" THEN "NOK"  
elseif [Fault Recovery Time]="N/A" or [Fault Recovery Time]=""" or isnull([Fault Recovery Time]) THEN "NOK"  
else "" endif
```

Aggregation Requirements

- Group data by Fault Level
- Use a Summarize Tool to compute:
 - Total PCMs
 - Within/Exceeded Target SLA
 - Within/Exceeded Hurdle SLA

SLA Percentage Metrics

- Achieved Target SLA % = [Within Target SLA] / [Total PCMs] * 100
- Achieved Hurdle SLA % = [Within Hurdle SLA] / [Total PCMs] * 100

Create both as FixedDecimal and V_WString

Join with SLA Benchmark Table

Join your results with the following benchmark table on Fault Level = Severity:

Severity	Target	Hurdle
-----	-----	-----
Emergency	90% < 2 hours	96% < 4 hours
Critical	90% < 4 hours	95% < 6 hours
Major	90% < 8 hours	95% < 12 hours
Minor	90% < 12 hours	95% < 24 hours

Sort output by the order of severity (# 1 to 4).

Final Output Columns

Ensure your output includes:

- Fault Level
- Total PCMs
- Within/Exceeded SLA (Target & Hurdle)
- Achieved SLA %
- Target and Hurdle benchmark strings

Deliverables

- Final Alteryx Workflow (.yxmd)
- Output Dataset (.xlsx or .csv)
- Screenshots showing key configuration steps